P5 a) CTV system $\dot{x}(t) = \begin{bmatrix} 0 & -t^2 \\ t^2 & -t^2 \end{bmatrix} x(t)$

$= A(t)\,x(t)$

$$A(t) = \begin{bmatrix} 0 & -t^2 \\ t^2 & -t^2 \end{bmatrix} = t^2 \underbrace{\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}}_{M} = t^2 M.$$

Here $A(t)$ is commutative, since for t_1 & t_2 two time instances, so, $A(t_1)\,A(t_2) = (t_1^2 M)(t_2^2 M) = (t_2^2 M)(t_1^2 M) = A(t_2)A(t_1)$

Formula for this commutative STM

$$\phi(t,\tau) = e^{\int_\tau^t A(s)\,ds} = e^{\int_\tau^t (s^2 ds)\,M}$$

$$\int_\tau^t s^2 ds = \left[\frac{s^3}{3}\right]_\tau^t = \frac{t^3}{3} - \frac{\tau^3}{3} = \frac{t^3 - \tau^3}{3} = u \text{ say}$$

Let $\therefore \quad \phi(t,\tau) = e^{(uM)}$ where $u = \frac{t^3 - \tau^3}{3}$

To Compute $e^{u.M}$, we get characteristic poly^n of M, to get eigen values of M

$\det(\lambda I - M) = 0$

$$\lambda I - M = \underbrace{\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}}_{\lambda I} - \underbrace{\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}}_{M} = \begin{bmatrix} \lambda & 1 \\ -1 & \lambda+1 \end{bmatrix}$$

$$\det(\lambda I - M) = \det\begin{bmatrix} \lambda & 1 \\ -1 & \lambda+1 \end{bmatrix} = \lambda(\lambda+1) + 1 = \lambda^2 + \lambda + 1$$

$\therefore$ characteristic eq^n $\chi_M = \lambda^2 + \lambda + 1 = 0$

$$\lambda = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm \sqrt{3}\,i}{2} \qquad i^2 = -1 \text{ here.}$$

$\therefore \lambda_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2} i \quad \& \quad \lambda_2 = -\frac{1}{2} - \frac{\sqrt{3}}{2} i$

$\therefore$ M is 2×2 $\quad M^2 + M + I = 0$

$\therefore$ M^2 & M & I are LI & we can write $e^{uM} = C_0 I + C_1 M$ ——①

Using λ_1

$$e^{u\left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i\right)} = C_0 + C_1\left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i\right)$$

or $e^{-\frac{u}{2}} e^{i\sqrt{3}/2 u} = C_0 + C_1\left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i\right)$

$$e^{-u/2}\left\{\cos\left(u\cdot\frac{\sqrt{3}}{2}\right) + i \sin\left(u\cdot\frac{\sqrt{3}}{2}\right)\right\} = C_0 + C_1\left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i\right)$$

$$e^{-u/2}\cos\left(u\frac{\sqrt{3}}{2}\right) + e^{-u/2}\cdot i \sin\left(u\frac{\sqrt{3}}{2}\right) = \left(C_0 - \frac{C_1}{2}\right) + i\left(C_1\frac{\sqrt{3}}{2}\right)$$

Comparing Re & Im parts

$$C_0 - \frac{C_1}{2} = e^{-u/2}\cos\left(\frac{\sqrt{3}}{2}u\right)$$

$$e^{-u/2}\sin\left(\frac{u\sqrt{3}}{2}\right) = C_1\frac{\sqrt{3}}{2}$$

$$\underline{\underline{C_1 = \frac{2}{\sqrt{3}} e^{-u/2}\sin\left(\frac{u\sqrt{3}}{2}\right)}}$$

$$\therefore C_0 = e^{-u/2}\cos\left(\frac{\sqrt{3}}{2}u\right) + \frac{1}{2}\frac{2}{\sqrt{3}} e^{-u/2}\sin\left(\frac{\sqrt{3}}{2}u\right)$$

$$\underline{\underline{C_0 = e^{-u/2}\left[\cos\left(\frac{\sqrt{3}}{2}u\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}u\right)\right]}}$$

plug C_0 & C_1 in ①,

$$C_0 I = \begin{bmatrix} C_0 & 0 \\ 0 & C_0 \end{bmatrix}, \quad C_1 M = \begin{bmatrix} 0 & -C_1 \\ C_1 & -C_1 \end{bmatrix}$$

$$\therefore e^{uM} = \begin{bmatrix} C_0 & 0 \\ 0 & C_0 \end{bmatrix} + \begin{bmatrix} 0 & -C_1 \\ C_1 & -C_1 \end{bmatrix} = \begin{bmatrix} C_0 & -C_1 \\ C_1 & C_0 - C_1 \end{bmatrix}$$

$$C_0 - C_1 = e^{-u/2}\left[\cos\left(\frac{\sqrt{3}}{2}u\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}u\right)\right] - \frac{2}{\sqrt{3}}e^{-u/2}\sin\left(\frac{\sqrt{3}}{2}u\right)$$

$$= e^{-u/2}\left[\cos\left(\frac{\sqrt{3}}{2}u\right) - \frac{1}{3}\sin\left(\frac{\sqrt{3}}{2}u\right)\right]$$

$$\therefore \quad \phi(t,\tau) = e^{uM} = \begin{bmatrix} C_0 & -C_1 \\ C_1 & C_0 - C_1 \end{bmatrix} \qquad \text{where, } u = \frac{t^3 - \tau^3}{6}$$

$$= e^{-u/2}\begin{bmatrix} \cos\left(\frac{\sqrt{3}}{2}u\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}u\right) & -\frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}u\right) \\ \frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}u\right) & \cos\left(\frac{\sqrt{3}}{2}u\right) - \frac{1}{3}\sin\left(\frac{\sqrt{3}}{2}u\right) \end{bmatrix}$$

$$= e^{-u/2}\begin{bmatrix} \cos\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) & -\frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) \\ \frac{2}{\sqrt{3}}\sin\left(\frac{t^3-\tau^3}{6}\right) & \cos\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}\,t^3-}{}\right. \end{bmatrix}$$

$$= e^{-\frac{t^3-\tau^3}{6}}\begin{bmatrix} \cos\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) & -\frac{2}{\sqrt{3}}\sin\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) \\ \frac{2}{\sqrt{3}}\sin\left(\frac{t^3-\tau^3}{6}\right) & \cos\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}(t^3-\tau^3)}{6}\right) \end{bmatrix}$$

Any

b) $x(2) = [1 \quad -1]^T$ to find $x(1)$.

$$x(t) = \phi(t,\tau)\,x(\tau)$$

$$x(1) = \phi(1,2)\cdot x(2)$$

$$u = \frac{t^3-\tau^3}{6} = \frac{1-8}{6} = -7/3 \quad \therefore \; e^{-u/2} = e^{7/6}$$

Also $\frac{\sqrt{3}}{2} u = \frac{\sqrt{3}}{2} \cdot -\frac{7}{3} = -\frac{7\sqrt{3}}{6}$

from 5(a),

$$\therefore \Phi(1,2) = e^{\frac{7}{6}} \begin{bmatrix} \cos\left(\frac{7\sqrt{3}}{6}\right) - \frac{1}{\sqrt{3}} \sin\left(\frac{7\sqrt{3}}{6}\right) & \frac{2}{\sqrt{3}} \sin\left(\frac{7\sqrt{3}}{6}\right) \\ -\frac{2}{\sqrt{3}} \sin\left(\frac{7\sqrt{3}}{6}\right) & \cos\left(\frac{7\sqrt{3}}{6}\right) + \frac{1}{\sqrt{3}} \sin\left(\frac{7\sqrt{3}}{6}\right) \end{bmatrix}$$

$$x(1) = \begin{bmatrix} \quad \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= e^{7/6} \begin{bmatrix} \cos\left(\frac{7\sqrt{3}}{6}\right) - \frac{1}{\sqrt{3}} \sin\left(\frac{7\sqrt{3}}{6}\right) - \frac{2}{\sqrt{3}} \sin\left(\frac{7\sqrt{3}}{6}\right) \\ -\frac{2}{\sqrt{3}} \sin\left(\frac{7\sqrt{3}}{6}\right) - \cos\left(\frac{7\sqrt{3}}{6}\right) - \frac{1}{\sqrt{3}} \sin\left(\frac{7\sqrt{3}}{6}\right) \end{bmatrix}$$

$$= e^{7/6} \begin{bmatrix} \cos\left(\frac{7\sqrt{3}}{6}\right) - \sqrt{3} \sin\left(\frac{7\sqrt{3}}{6}\right) \\ -\cos\left(\frac{7\sqrt{3}}{6}\right) - \sqrt{3} \sin\left(\frac{7\sqrt{3}}{6}\right) \end{bmatrix}$$ Ans

— x —

PG LTV

$$\dot{x}(t) = \begin{bmatrix} -1 & 0 \\ t & -2 \end{bmatrix} x(t)$$

Lower triangular.

here $A(t_1)A(t_2) \neq A(t_2)A(t_1)$ so non commutative

special case 1

Let $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$

$$\dot{x}_1(t) = -x_1(t) \quad \text{—①}$$

$$\dot{x}_2(t) = t\,x_1(t) - 2x_2(t) \quad \text{—②}$$

from ① $\frac{dx_1}{dt} = -x_1$ or $\int \frac{1}{x_1} dx_1 = \int -dt$

or $\ln|x_1| = -t + c$

$|x_1| = e^{-t+c}$

$= e^c e^{-t} = c_1 e^{-t}$

$\therefore \quad x_1(t) = + C_1 e^{-t}$ ——③ or $x_1(t) = e^{-t} x_1(0)$

③ into ②

$$\dot{x}_2(t) = t\, x_1(0) e^{-t} - 2 x_2(t)$$

$$\dot{x}_2(t) + 2x_2(t) = x_1(0) t e^{-t}$$

Use integrating factor $\mu(t) = e^{\int p(t)\,dt}$ where $p(t) = 2$

$= e^{2\int dt} = e^{2t}$

multiply both sides

$$e^{2t}\dot{x}_2(t) + 2e^{2t} x_2(t) = t e^{-t} \cdot e^{2t} x_1(0) = t e^{t} x_1(0)$$

$$\therefore \frac{d}{dt}\left[e^{2t} x_2(t) \right] = t e^{t} x_1(0)$$

$$\int_0^t d\left[e^{2t} x_2(t) \right] = \int_0^t t e^{t} x_1(0)\, dt$$

$e^{2t} x_2(t) - x_1(0)$ [illegible]

integrating from 0 to t

$$\text{or} \quad \left[e^{2t} x_2(t) \right]_0^t = x_1(0) \left[t e^{t} - e^{t} \right]_0^t$$

$$\text{or} \quad e^{2t} x_2(t) - x_2(0) = \left[t e^{t} - e^{t} + 1 \right] x_1(0)$$

$$e^{2t} x_2(t) = \left[(t-1) e^{t} + 1 \right] x_1(0) + x_2(0)$$

$$x_2(t) = \left[(t-1) e^{-t} + e^{-2t} \right] x_1(0) + e^{-2t} x_2(0)$$ ——④

by parts

$\int t e^{t} dt$

$= t e^{t} - \int e^{t} dt$

$= t e^{t} - e^{t} + c$

$=$

$\therefore \quad x(t) = \Phi(t)\, x(0)$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} e^{-t} & 0 \\ (t-1)e^{-t} + e^{-2t} & e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

↑ Φ fundamental matrix

Ans

State transition matrix

$$\phi(t,\tau) = \phi(t)\,\phi^{-1}(\tau)$$

$$\det \phi(\tau) = \begin{vmatrix} e^{-\tau} & 0 \\ (\tau-1)e^{-\tau} & e^{-2\tau} \end{vmatrix} = e^{-\tau}\cdot e^{-2\tau} - 0 = e^{-3\tau}$$

$$\& \quad \phi^{-1}(\tau) = \frac{1}{e^{-3\tau}} \begin{bmatrix} e^{-2\tau} & 0 \\ -(\tau-1)e^{-\tau} & e^{-\tau} \end{bmatrix} = \begin{bmatrix} e^{\tau} & 0 \\ -(\tau-1)e^{2\tau} & e^{2\tau} \end{bmatrix}$$

$$\phi(t,\tau) = \begin{bmatrix} e^{-t} & 0 \\ (t-1)e^{-t} & e^{-2t} \end{bmatrix} \begin{bmatrix} e^{\tau} & 0 \\ -(\tau-1)e^{2\tau} & e^{2\tau} \end{bmatrix}$$

$$= \begin{bmatrix} e^{-t}\cdot e^{\tau} + 0 & 0+0 \\ (t-1)e^{-t}e^{\tau} + e^{-2t}(1-\tau)e^{2\tau} & 0 + e^{-2t}e^{2\tau} \end{bmatrix}$$

$$\Phi(t,\tau) = \begin{bmatrix} e^{-(t-\tau)} & 0 \\ (t-1)e^{-(t-\tau)} + (1-\tau)e^{-2(t-\tau)} & e^{-2(t-\tau)} \end{bmatrix}$$ Ans

—x—

P3 $$A = \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}}_{T} \underbrace{\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}}_{J} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}^{-1}}_{T^{-1}}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix}$$

$\therefore A = TJT^{-1}$

Jordan Canon form

a) CT LTI system $\dot{x} = Ax$ stability

$$J = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$J_1 = [-1]_{1\times 1}$, $\lambda_1 = -1$, $v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$J_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}_{2\times 2}$, $\lambda_2 = 0$, $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$\lambda_2, \lambda_3 = 0$, AM = 2 $\lambda_3 = 0$, $v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

```
clc;
clear;

% problem 1(b)

A = [1/3  1/3  1/3  0;
    1/3  1/3  1/3  0;
    1/4  1/4  1/4  1/4;
    0    0    1/2  1/2];

% Right Eigenvectors and Eigenvalues
[V,D] = eig(A);

fprintf('Eigenvalues:\n');
```

```
Eigenvalues:
```

```
disp(diag(D));
```

```
    1.0000
    0.5643
    0.0000
   -0.1477
```

```
fprintf('Right eigenvectors:\n');
```

```
Right eigenvectors:
```

```
disp(V);
```

```
    0.5000    0.3584   -0.7071   -0.2946
    0.5000    0.3584    0.7071   -0.2946
    0.5000   -0.1100    0.0000    0.7196
    0.5000   -0.8550    0.0000   -0.5555
```

```
% Left Eigenvectors
W = inv(V);
fprintf('Left eigenvectors:\n');
```

```
Left eigenvectors:
```

```
disp(W);
```

```
    0.5000    0.5000    0.6667    0.3333
    0.4713    0.4713   -0.1929   -0.7497
   -0.7071    0.7071   -0.0000   -0.0000
   -0.2754   -0.2754    0.8969   -0.3462
```

$$x(0) = c_1 v_1 + c_2 v_2 + c_3 v_3 \quad \text{or} \quad x(0) = T \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$\text{or} \quad \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = T^{-1} x(0)$$

$$T^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix} \quad x(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix}$$

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix}$$

$$c_1 = x_3(0) = x_3(0) \quad \text{—①}$$

$$c_2 = x_2(0) - x_3(0) \quad \text{—②}$$

$$c_3 = x_1(0) - x_2(0) \quad \text{—③}$$

a) Since $\lambda_1 = -1$, & $\lambda_2 = 0$ from 2×2 Jordan block, it's defective.

① No eigen value in open right half of plane $\mathbb{C}$.

② $\lambda = 0$ is on jω axis & from J_2 with size 1.

so $\lambda = 0$ defective

& CT LTI system is unstable. Answer

General solution here

$$x(t) = e^{At} x(0)$$

$$A^k = T J^k T^{-1}$$

$$\& \; e^{At} = T e^{Jt} T^{-1}$$

$$\therefore x(t) = T e^{Jt} T^{-1} x(0) \qquad \therefore e^{Jt} T^{-1} x(0)$$

$$e^{Jt} = \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} e^{-t} & 0 & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_1 e^{-t} \\ c_2 + c_3 t \\ c_3 \end{bmatrix}$$

$$\therefore x(t) = c_1 e^{-t} \underbrace{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}}_{v_1} + (c_2 + c_3 t) \underbrace{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}_{v_2} + c_3 \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{v_3} \quad \text{—④}$$

(i) Yes if $x(t)$ converges to 0 as $t \to \infty$ for (4),

$c_1 e^{-t} v_1 \to 0$ but if $c_2 v_2 \neq 0$ & $c_3 \neq 0$

since $e^{-t} \to 0$ & $c_3 t v_2 \neq 0$

second term and third term do not goto zero.

$\therefore$ to make $x(t) \to 0$ as $t \to \infty$, $c_2 = 0$ & $c_3 = 0$

& $c_1 \neq 0$ so let $x(0) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ non zero $\therefore c_1 = x_3(0) = 1$

$c_2 = 0$ & $c_3 = 0$] from (1)(2) & (3)

So Yes, $x(t) = e^{-t} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ will tend to $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ when $t \to \infty$

Answer

(ii) Yes if $x(t)$ converge to some $x(\infty) \neq 0$ as $t \to \infty$

from eqn (4), $c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \to 0$ since $e^{-t} \to 0$ (v_1)

2nd term $\to$ $c_3 t v_2$ but as $t \to \infty$, if $c_3 \neq 0$ this terms become unbounded / blows up to infinity

So $c_3 = 0$

i.e. $\lim_{t \to \infty} x(t) = \lim_{t \to \infty} \left(c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = 0 + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$= c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$c_2 \neq 0$ if limit $t \to \infty$ is non zero

let $x(0) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ $\therefore$ from (1), (2) & (3)

$x(t) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$c_1 = x_3(0) = 0$

$c_2 = x_2(0) - x_3(0) = 1 - 0$

$c_3 = x_1(0) - x_2(0) = 1 - 1 = 0$

$x(t) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ So Yes, $\lim_{x \to \infty} x(t) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \neq 0$

Answer.

(iii) from (4), as $t \to \infty$, $c_3 t v_2 \to \infty$ if $c_3 \neq 0$

we choose $x(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $c_1 = 0$, $c_2 = 0$ } from (1), (2) (3)

& $c_3 = 1 - 0$

$\therefore x(t) = t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ as $t \to \infty$ $\|x(t)\| \to \infty$ Answer

iv) Bounded no limit

can't have grow term $c_3 t v_2$

if $c_3 = 0$, $x(t) = c_1 e^{-t}\begin{bmatrix}1\\1\end{bmatrix} + c_2\begin{bmatrix}1\\0\end{bmatrix}$

as $\lim_{t\to\infty}(x(t)) = \lim_{t\to\infty}\left(c_1 e^{-t}\begin{bmatrix}1\\1\end{bmatrix} + c_2\begin{bmatrix}1\\0\end{bmatrix}\right)$ ($e^{-t} \to 0$)

$$x(t) \to c_2\begin{bmatrix}1\\0\end{bmatrix}$$

so bounded and have limit.

So, No, bounded but not have limit not possible. Ans

(b) DT LTI system $x[k+1] = Ax[k]$ Stability

Check 1) A has no eigen value outside the close unit disk of $\mathbb{C}$

so $|\lambda_i| \leq 1$

& check 2) A has eigen val on the unit circle $\mathbb{C}$ each non defective so each $|\lambda_i| = 1$ is non defective.

here three eigen values are -1, 0 & 0

$|-1| = 1$ from J_1 1×1 J-block so non defective.

$|0| = 0 < 1$ from 2×2 J block but lies inside unit disk

Because no eigen value lie outside close unit disk of $\mathbb{C}$ ($|\lambda| \leq 1$) and eigen value on the unit circle is non defective ($\lambda_1 = -1$); the overall DT LTI system is 'Marginally stable'.

$x[k+1] = Ax[k]$

$\therefore x[k] = A^k x[0]$

$A = TJT^{-1}$

$A^k = TJ^kT^{-1}$

$\therefore x[k] = TJ^kT^{-1}x[0]$

Since $J = \begin{bmatrix}-1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{bmatrix}$ $\therefore J^k = \begin{bmatrix}(-1)^k & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$, $k \geq 2$

$\therefore x[k] = T J^k \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$ where $T^{-1}x[0] = [c_1\ c_2\ c_3]^T$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} (-1)^k & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} (-1)^k c_1 \\ 0 \\ 0 \end{bmatrix} = (-1)^k c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{---- (2)}$$

$\therefore x[k] = (-1)^k c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ for $k \geq 2$

Case(i) To find $x[0] \neq 0$ as $k \to \infty$

i.e. the term $(-1)^k c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \neq 0$ unless $c_1 = 0$

we got in page (7), $c_1 = x_3[0]$ $\therefore x_3[0] = 0$

$\therefore$ The condition for convergence to zero as $k \to \infty$ is

$c_1 = x_3[0] = 0$

pick $x[0] = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ i.e. v_2 | v_3 should also work

$\therefore c_1 = 0$, $c_2 = x_2(0) - x_3(0) = 1 - 0 = 1$, $c_3 = x_1(0) - x_2(0) = 1 - 1 = 0$

$\therefore$ let $x[k] = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, $k \geq 2 \to 0$

$\therefore$ Yes $x[0]$ can be $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ [or could be $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ too] or many other choices!

Answer

(ii) Converges to finite $x[\infty] \neq 0$.

if $c_1 = 0$, $x[k] = 0$ for $k \geq 2$ $x[k] \to 0$ so not nonzero.

if $c_1 \neq 0$ since $(-1)^k$ either -1 or 1,

$x[k]$ alternate $c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ or $-c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ or oscillates. so No nonzero initial condition found

Answer

(iii) is unbounded as $k \to \infty$?

$$x[k] = c_1 (-1)^k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad |(-1)^k| = 1$$

$$\| x[k] \| = |c_1| \left\| \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\| = |c_1| \sqrt{3} = \text{const}$$

so No, can't remain unbounded Answer

(iv) Bounded but does not have limits

$c_1 \times x_3 \neq 0$ is needed but we choose $x(0) = \begin{bmatrix} v_1 \\ 1 \\ 1 \end{bmatrix}$

$c_1 = x_3[0] = 1$

$\therefore x[k] = (-1)^k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ for $k \geq 0$ a vector can

oscillate as $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ & $\begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$

so Yes. Answer

—×—

(P4) DT LTI

$$x[k+1] = Ax[k] = \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}_{T} \underbrace{\begin{bmatrix} -0.5 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}}_{J} \underbrace{\begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{T^{-1}} x[k] \qquad k = 0, 1, \ldots$$

Eigenvalue $\lambda_1 = -0.5$ from 1×1 Jordan $J_1 = [-0.5]$

right eigenvector $v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

left eigenvector $w_1^T = [1 \;\; -1 \;\; 0]$

$\lambda_2 = -1$ from 2×2 $J_2 = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$ AM = 2, GM = 1

right eigenvectors $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ & $v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

(a) To check stability

$|\lambda_1| = |-0.5| = 0.5 < 1$, inside open disk

$|\lambda_2| = |-1| = 1$, lies on unit disk.

$\lambda_2 = -1$ from 2×2 $J_2 \Rightarrow$ GM < AM $\therefore \lambda_2 = -1$ defective.

DT LTI system, an eigen value on the unit circle must be non defective for stability, except,

Here system is unstable.

$J_2^k = \begin{bmatrix} (-1)^k & k(-1)^{k-1} \\ 0 & (-1)^k \end{bmatrix}$, the element $k(-1)^{k-1}$ as k is large as $k \to \infty$, $k(-1)^{k-1}$ is large so system is unstable & unbounded

(b) $\therefore J^k = \begin{bmatrix} (-0.5)^k & 0 & 0 \\ 0 & (-1)^k & k(-1)^{k-1} \\ 0 & 0 & (-1)^k \end{bmatrix}$

Col^r of T $v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

from TJ^k,

mode $(\lambda_1 = -0.5)$ is $(-0.5)^k v_1 = (-0.5)^k \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ ✓

mode $(\lambda_2 = -1)$ is $(-1)^k v_2 = (-1)^k \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ ✓ ← 3 modes

mode $(\lambda_2 = -1)$ is $k(-1)^{k-1} v_2 + (-1)^k v_3$

$= k(-1)^{k-1} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + (-1)^k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ ✓

Ans

$x[k] = TJ^k T^{-1} x[0]$

$T^{-1} x[0] = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$ & $x[0] = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$\therefore \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$c_1 = x_1 - x_2$ & $c_3 = x_3$

$c_2 = x_2 - x_3$

$\therefore J^k \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} (-0.5)^k & 0 & 0 \\ 0 & (-1)^k & k(-1)^{k-1} \\ 0 & 0 & (-1)^k \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_1(-0.5)^k \\ c_2(-1)^k + c_3 k(-1)^{k-1} \\ c_3(-1)^k \end{bmatrix}$

$$\Rightarrow \quad TJ^kT^{-1}\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1(-0.5)^k \\ c_2(-1)^k + c_3 k(-1)^{k-1} \\ c_3(-1)^k \end{bmatrix}$$

$$x[k] = A^k x = TJ^kT^{-1}x[0] = c_1(-0.5)^k \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \left[c_2(-1)^k + c_3 k(-1)^{k-1}\right]\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{---} ①$$

$$+ \quad c_3(-1)^k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Answer ?

$$x[k] = ① (x_1 - x_2)(-0.5)^k \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (x_2 - x_3)(-1)^k \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$+ ② x_3 \left(k(-1)^{k-1}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + (-1)^k \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right)$$

Answer ?

here $\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $x[0]$ & both equations are expressed as c's & one with x's

(c) $V \in \mathbb{R}^3$ s.t $x[0] \in V \Rightarrow x[k] \to 0$ as $k \to \infty$.

what ? ∴

eq ⁼ ① in page 13 here,

first term as $k \to \infty$ $(-0.5)^k \to 0$, $|-0.5| < 1$ ∴ c_1 stays

c_1 term decay to zero eventually.

second c_2 term, $(-1)^k$, $|-1| = 1$ magnitude same

so only oscillates & does not decay.

∴ c_2 needs to be $c_2 = 0$

third c_3 term,

$k(-1)^{k-1}$ grows linearly as k, and can get unbounded as $k \to \infty$

$(-1)^k$ oscillates as before

∴ c_3 needs to be $c_3 = 0$

$c_3 = x_3 = 0$, $c_2 = x_2 - 0$ ∴ $x_2 = c_2 = 0$, $x_2 = 0$,

$$V = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\right\} = \left\{\begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^3 \middle| \begin{array}{l} \text{with} \\ x_1 \in \mathbb{R} \end{array}\right\} \quad \dim(V) = 1$$

Answer

(d) $W \in \mathbb{R}^3$ s.t $x[0] \in W$, $\|x[k]\|$ doesn't diverge to ∞ as $k \to \infty$?

c_1 term $(-0.5)^k \to 0$ gets smaller as $k \to \infty$ so c_1 stays bounded.

c_2 term $|(-1)^k| = 1$ but flips sign but magnitude is always 1. So bounded. c_2 stays.

c_3 term $k(-1)^{k+1}$ it grows as $k \to \infty$ and will get unbounded so c_3 needs to be $c_3 = 0$

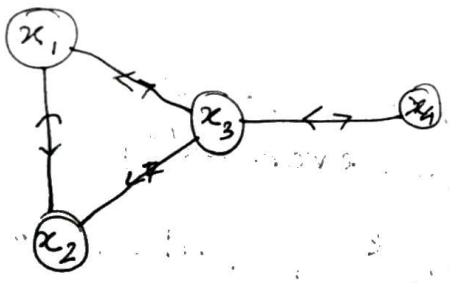
$c_3 = x_3 = 0 \Rightarrow (x_1 - x_2)v_1 + x_2 v_2$

$\therefore$ or, $x[0] = c_1 v_1 + c_2 v_2$

so $W = \text{span}\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}\right\} \Rightarrow \{v_1, v_2 \in \mathbb{R}^3\}$

dimension of $W = 2$ Answer

P1



sensor	Connection
1	2, 3
2	1, 3
3	1, 2, 4
4	3

(a) $x[k] = \begin{bmatrix} x_1[k] & x_2[k] & x_3[k] & x_4[k] \end{bmatrix}^T \in \mathbb{R}^4$

To find DT LTI $x[k+1] = Ax[k]$ $A \in \mathbb{R}^{4\times 4}$

algo says,

each sensor averages itself + neighbors

Given, $x_2[k+1] = \dfrac{x_1[k] + x_3[k] + x_2[k]}{3}$

$\therefore$ $x_1[k+1] = \dfrac{x_1[k] + x_2[k] + x_3[k]}{3}$ & $x_4[k+1] = \frac{1}{2}(x_3[k] + x_4[k])$

$x_3[k+1] = \dfrac{x_1[k] + x_2[k] + x_3[k] + x_4[k]}{4}$

$\therefore \quad x[k+1] = Ax[k]$ where $x[k] = \begin{bmatrix} x_1[k] \\ x_2[k] \\ x_3[k] \\ x_4[k] \end{bmatrix}$

$$A = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 1/3 & 1/3 & 1/3 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}$$

Ans

(b) eigen vals & left & right eigen vects.

Used Matlab here (Attached) $A v_i = \lambda v_i$

$\lambda_1 = 1.0000$

$\lambda_2 = 0.5643$

$\lambda_3 = 0.0000$

$\lambda_4 = -0.1477$

Ans

$\lambda_1 = 1$, right eigen vector $v_1 = \begin{bmatrix} 0.5000 \\ 0.5000 \\ 0.5000 \\ 0.5000 \end{bmatrix}$

$\lambda_2 = 0.5643$, right eigen vector $v_2 = \begin{bmatrix} 0.3584 \\ 0.3584 \\ -0.1100 \\ -0.8550 \end{bmatrix}$

$\lambda_3 = 0$, right eigen vector $v_3 = \begin{bmatrix} -0.7071 \\ 0.7071 \\ 0.0000 \\ 0.0000 \end{bmatrix}$

Ans

$\lambda_4 = -0.1477$, right eigen vector $v_4 = \begin{bmatrix} -0.2946 \\ -0.2946 \\ 0.7196 \\ -0.5555 \end{bmatrix}$

left eigen vector $w^T v_i = \lambda w_i^T$

~~$\lambda_1 = 1$, $w_1^T = [0.5000 \quad 0.4713 \quad -0.7071 \quad -0.2754]$~~

~~$\lambda_2 = 0.5643$, $w_2^T = [0.5000 \quad 0.4713 \quad 0.7071 \quad -0.2754]$~~

$\therefore \lambda_1 = 1$, $w_1^T = [0.50000 \quad 0.50000 \quad 0.6667 \quad 0.3333]$

$\lambda_2 = 0.5643$, $w_2^T = [0.4713 \quad 0.4713 \quad -0.1929 \quad -0.7497]$

$\lambda_3 = 0$, $w_3^T = [-0.7071 \quad 0.7071 \quad 0 \quad 0]$

$\lambda = -0.1477$, $w_4^T = [-0.2754 \quad -0.2754 \quad 0.8969 \quad -0.3462]$

Ans

$A = V D V^{-1} = \sum_{i=1}^{4} \lambda_i v_i w_i^T$ A is diagonalizable.

$$x[k] = A^k x[0] = \sum_{i=1}^{4} \lambda_i^k v_i w_i^T x[0]$$

$$x[k] = \left[1^k v_1 (w_1^T x[0]) + (0.5643)^k v_2 (w_2^T x[0]) + (0)^k v_3 (w_3^T x[0]) + (-0.1477)^k v_4 (w_4^T x[0])\right]$$

(1)

[Answer]

w_i^T & v_i are defined in previous page with their numerical value.

(c) $w^T = [\,1/4 \quad 1/4 \quad 1/3 \quad 1/6\,] = [\,0.$

$$x[k+1] = A x[k]$$

$$w^T x[k+1] = w^T A x[k]$$

$$= w^T x[k]$$

w^T here is for $\lambda_1 = 1$.

$\therefore$ since $w^T x[k+1] = w^T x[k]$

$$\therefore w^T x[k] = w^T x[0] = \bar{x}, \quad k = 0, 1, 2 \ldots$$

$$\therefore w^T x[k] = \begin{bmatrix} 1/4 & 1/4 & 1/3 & 1/6 \end{bmatrix} \begin{bmatrix} x_1[k] \\ x_2[k] \\ x_3[k] \\ x_4[k] \end{bmatrix}$$

$$\therefore \frac{1}{4} x_1[k] + \frac{1}{4} x_2[k] + \frac{1}{3} x_3[k] + \frac{1}{6} x_4[k]$$

$$= \frac{1}{4} x_1[0] + \frac{1}{4} x_2[0] + \frac{1}{3} x_3[0] + \frac{1}{6} x_4[0]$$

$$\therefore \bar{x} = \frac{1}{4} x_1[0] + \frac{1}{4} x_2[0] + \frac{1}{3} x_3[0] + \frac{1}{6} x_4[0]$$

$\therefore$ by induction $w^T x[k] = \bar{x}$ const for $\forall k$.

[Answer]

(d) $e[k] = x[k] - \bar{x}\,[1\ 1\ 1\ 1]^T$ ——— (1)

From eqn (1) in page 16, getting $x[k]$ here

$$e[k] = v_1 w_1^T x[0] + (0.5643)^k v_2 (w_2^T x[0]) + 0^k v_3 (w_3^T x[0]) + (-0.1477)^k v_4 (w_4^T x[0]) - v_1 (w_1^T \eta[0])$$

$\eta[0] \rightarrow$ This is $\bar{x}\begin{bmatrix}1\\1\\1\\1\end{bmatrix}$

$$e[k] = (0.5643)^k v_2 (w_2^T x[0]) + 0^k v_3 (w_3^T x[0]) + (-0.1477)^k v_4 (w_4^T x[0])$$

$|0.5643| < 1$, $|-0.1477| < 1$ & $(-0.1477)^k \to 0$ as $k \to \infty$

$(0.5643)^k \to 0$ & $(-0.1477)^k \to 0$ as $k \to \infty$

ok so now, as $k \to \infty$ since each term $\to 0$

$$\lim_{k\to\infty} e[k] = 0$$

$$\therefore\ x[k] \to \bar{x}\begin{bmatrix}1\\1\\1\\1\end{bmatrix} \quad \text{or} \quad \lim_{k\to\infty} x[k] = \bar{x}\begin{bmatrix}1\\1\\1\\1\end{bmatrix}$$

↑ from (1),

ie., $x_1[k], x_2[k], x_3[k], x_4[k] \to \bar{x}$

This proves consensus can be achieved asymptotically & its value is $\bar{x}$.

$$\bar{x} = \frac{1}{4} x_1[0] + \frac{1}{4} x_2[0] + \frac{1}{3} x_3[0] + \frac{1}{6} x_4[0]$$

Answer

(e)

$$e[k] = (0.5643)^k v_2 (w_2^T x[0]) + 0^k v_3 (w_3^T x[0]) + (-0.1477)^k v_4 (w_4^T x[0])$$

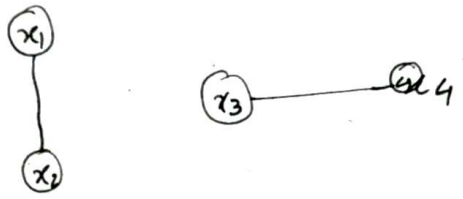
largest $\lambda_2 = 0.5643$ absolute value → so slowest decay

Since, $|0.5643| = 0.5643$, $|-0.1477| = 0.1477$ & $|0| = 0$

$\therefore \alpha = 0.5643$ here

$\therefore e[k] \to 0$ as $k \to \infty$ due to dominant rate given by $(0.5643)^k$ Ans

(f)



$$x_1[k+1] = \frac{x_1[k] + x_2[k]}{2} \qquad x_2[k+1] = \frac{x_2[k] + x_1[k]}{2}$$

$$x_3[k+1] = \frac{x_3[k] + x_4[k]}{2} \qquad \& \; x_4[k+1] = \frac{x_3[k] + x_4[k]}{2}$$

$$A_{dis} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix}$$

Answer.

eigen values. $J^{2\times 2} B = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$

$$\det(B - \lambda I) = 0 \quad \begin{vmatrix} 1/2 - \lambda & 1/2 \\ 1/2 & 1/2 - \lambda \end{vmatrix} = 0$$

$$(1/2 - \lambda)^2 - \frac{1}{4} = 0 \quad \Rightarrow \frac{1}{4} - \lambda + \lambda^2 - \frac{1}{4} = 0$$

$$\text{or } \lambda^2(\lambda - 1) = 0 \quad \therefore \lambda = 0, \; \lambda = 1$$

two identical Jordan blocks

so four eigen values are $\lambda_1 = \lambda_2 = 0$ & $\lambda_3 = \lambda_4 = 1$.

$AM = 2$ $AM = 2$

Answer

$\lambda_3 = \lambda_4 = 1$

$$\begin{bmatrix} -1/2 & 1/2 & 0 & 0 \\ 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & -1/2 & 1/2 \\ 0 & 0 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-\frac{1}{2}v_1 + \frac{1}{2}v_2 = 0 \qquad v_1 = v_2$$

$$-\frac{1}{2}v_3 + \frac{1}{2}v_4 = 0 \qquad \therefore v_3 = v_4$$

$$\therefore v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore$ for 2 eigen space defined by 1, is spanned by $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

$\lambda_1 = \lambda_2 = 0$

$$\begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & -1/2 & 1/2 \\ 0 & 0 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$v_1 = -v_2, \quad v_3 = -v_4$$

$$\therefore v_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

$\therefore$ eigen space for 0 spanned by $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}$

$\therefore$ right eigen vectors

$$\hookrightarrow \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix}$$
$$\begin{bmatrix} -1 & 0 & +1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & +1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Ans

left eigenvector A_{dis}^T = Adis since symmetric.

$$w_1^T = [-\tfrac{1}{2} \;\; \tfrac{1}{2} \;\; 0 \;\; 0]$$

$$w_2^T = [0 \;\; 0 \;\; -\tfrac{1}{2} \;\; \tfrac{1}{2}]$$

$$w_3^T = [\tfrac{1}{2} \;\; \tfrac{1}{2} \;\; 0 \;\; 0]$$

$$w_4^T [0 \;\; 0 \;\; \tfrac{1}{2} \;\; \tfrac{1}{2}]$$

left eigenvectors $\hookrightarrow$ $\begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ Answer

$$\therefore \quad x[0] = \begin{bmatrix} x_1[0] \\ x_2[0] \\ x_3[0] \\ x_4[0] \end{bmatrix}$$

$$w_1^T x_1[0] = \frac{-x_1[0] + x_2[0]}{2}, \qquad w_2^T x_2[0] = \frac{-x_3[0] + x_4[0]}{2}$$

$$w_3^T x_3[0] = \frac{x_1[0] + x_2[0]}{2}, \qquad w_4^T x_4[0] = \frac{[x_3[0] + x_4[0]}{2}$$

disconnected system,

$$\therefore x[k] = 0^k v_1 \left(\frac{-x_1[0] + x_2[0]}{2}\right) + 0^k v_2 \left(\frac{-x_3[0] + x_4[0]}{2}\right)$$

$$+ 1^k v_3 \left[\frac{x_1[0] + x_2[0]}{2}\right] + 1^k v_4 \left(\frac{x_3[0] + x_4[0]}{2}\right)$$

$$\therefore x[k] = \frac{x_1[0] + x_2[0]}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \frac{x_3[0] + x_4[0]}{2} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore x[k] = A^k x[0] = \sum_{i=1}^{4} \lambda_i^k v_i w_i^T x[0]$$

$$x[k] = \begin{bmatrix} (x_1[0] + x_2[0])/2 \\ (x_1[0] + x_2[0])/2 \\ (x_3[0] + x_4[0])/2 \\ (x_3[0] + x_4[0])/2 \end{bmatrix}, \quad k \geq 1$$

Answer

(g) $x[k] \longrightarrow \bar{x}\begin{bmatrix}1\\1\\1\\1\end{bmatrix}$

from eqn ① in page 20,

$$x[k] = \frac{x_1[0]+x_2[0]}{2}\begin{bmatrix}1\\1\\0\\0\end{bmatrix} + \frac{x_3[0]+x_4[0]}{2}\begin{bmatrix}0\\0\\1\\1\end{bmatrix}$$

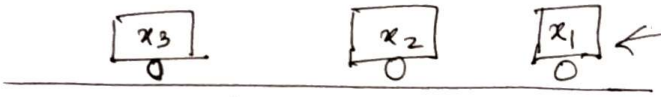
$$= \bar{x}_{12}\begin{bmatrix}1\\1\\0\\0\end{bmatrix} + \bar{x}_{34}\begin{bmatrix}0\\0\\1\\1\end{bmatrix} \qquad \left|\begin{array}{l}\bar{x}_{12} = \frac{1+1}{2} = 1\\ \bar{x}_{34} = 0.\end{array}\right\}$$

But $\bar{x}_{12} \neq \bar{x}_{34}$

and if $x = \begin{bmatrix}1\\1\\0\\0\end{bmatrix}$, $\bar{x}_{12} = 1$ & $\bar{x}_{34} = 0$ so they are not converging to single value. TI, d.r. Not possible & Justified.

Ans.

(P2)



$x_1[0]$, $x_2(0)$, $x_3(0)$ at $t = 0$

dynamics

$$\dot{x}_1(t) = x_2(t) - x_1(t)$$

$$\dot{x}_2(t) = \frac{x_1(t)+x_3(t)}{2} - x_2(t)$$

$$\dot{x}_3(t) = x_2(t) - x_3(t)$$

$$\therefore A = \begin{array}{c}\begin{matrix}x_1 & x_2 & x_3\end{matrix}\\ \begin{bmatrix}-1 & 1 & 0\\ \frac{1}{2} & -1 & \frac{1}{2}\\ 0 & 1 & -1\end{bmatrix}\end{array}$$

This is the CT LTI system where $\dot{x}(t) = Ax(t)$

To find eigen vals we solve $\det(A - \lambda I) = 0$

$$\begin{vmatrix}-1-\lambda & 1 & 0\\ \frac{1}{2} & -1-\lambda & \frac{1}{2}\\ 0 & 1 & -1-\frac{1}{2}\end{vmatrix} = 0$$

$$(-1-\lambda)\begin{vmatrix}(-1-\lambda) & 1/2\\ 1 & (-1-\lambda)\end{vmatrix} - \begin{vmatrix}\frac{1}{2} & 1/2\\ 0 & -1-\lambda\end{vmatrix} = 0$$

$$(-1-\lambda)\left\{(-1-\lambda)^2 - \tfrac{1}{2}\right\} - \tfrac{1}{2}(-1-\lambda) = 0$$

$$= (-1-\lambda)\left[(-1-\lambda)^2 - \tfrac{1}{2} - \tfrac{1}{2}\right] = (-1-\lambda)\left[(-1-\lambda)^2 - 1\right] = 0$$

$$= (-1-\lambda)\left[(-1-\lambda)(-1-\lambda) - 1\right] = 0$$

$$= (-\lambda)(-1-\lambda)(-\lambda-2) = 0$$

$\therefore \lambda_1 = 0, \lambda_2 = -1$ & $\lambda_3 = -2$ $\therefore$ eigenvalues are $0, -1, -2$

$\underbrace{\qquad}_{AM=2}$

$\lambda_1 = 0$,

$$\begin{bmatrix}-1 & 1 & 0\\ 1/2 & -1 & 1/2\\ 0 & 1 & -1\end{bmatrix}\begin{bmatrix}\tilde{v}_1\\ \tilde{v}_2\\ \tilde{v}_3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$-\tilde{v}_1 + \tilde{v}_2 = 0 \qquad \tilde{v}_2 - \tilde{v}_3 = 0 \qquad v_1 = \begin{bmatrix}1\\1\\1\end{bmatrix}$

$\tilde{v}_1 = \tilde{v}_2 \qquad \tilde{v}_2 = \tilde{v}_3$

$\lambda_2 = -1$

$$\begin{bmatrix}0 & -1 & 0\\ -1/2 & 0 & -1/2\\ 0 & -1 & 0\end{bmatrix}\begin{bmatrix}\tilde{v}_1\\ \tilde{v}_2\\ \tilde{v}_3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$\tilde{v}_2 = 0 \qquad \tilde{v}_1 + \tilde{v}_3 = 0 \qquad \tilde{v}_2 = 0 \qquad \therefore v_2 = \begin{bmatrix}-1\\0\\1\end{bmatrix}$

$\tilde{v}_1 = -v_3$

$\lambda_3 = -2$

$$\begin{bmatrix}-1 & -1 & 0\\ -1/2 & -1 & -1/2\\ 0 & -1 & -1\end{bmatrix}\begin{bmatrix}\tilde{v}_1\\ \tilde{v}_2\\ \tilde{v}_3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$v_3 = \begin{bmatrix}1\\-1\\1\end{bmatrix}$

$v_1 = -v_2 \qquad v_2 = -v_3$

$$\therefore \quad T = [v_1\ v_2\ v_3] = \begin{bmatrix}1 & -1 & 1\\ 1 & 0 & -1\\ 1 & 1 & 1\end{bmatrix}$$

$$T^{-1} = \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

$R_2 \to R_2 - R_1$ ↓ $R_3 \to R_3 - R_1$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & -1 & 1 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 \end{array}\right]$$

↓ $R_3 \to R_3 - 2R_2$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & -1 & 1 & 0 \\ 0 & 0 & 4 & 1 & -2 & 1 \end{array}\right]$$

$R_2 \to R_2 + 2R_3$ ↓ $R_3 \to \frac{1}{4} R_3$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 1/4 \end{array}\right]$$

↓ $R_1 \to R_1 - R_3$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 3/4 & 1/2 & -1/4 \\ 0 & 1 & 0 & -1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 1/4 \end{array}\right]$$

↓ $R_1 \to R_1 + R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/4 & 1/2 & 1/4 \\ 0 & 1 & 0 & -1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 1/4 \end{array}\right] \quad T^{-1} \uparrow$$

$$\therefore A = T \Lambda T^{-1} = \underbrace{\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix}}_{T} \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}}_{\Lambda} \underbrace{\begin{bmatrix} 1/4 & 1/2 & 1/4 \\ -1/2 & 0 & 1/2 \\ 1/4 & -1/2 & 1/4 \end{bmatrix}}_{T^{-1}}$$

for CT LTI system $\dot{x}(t) = e^{At} x(0)$

$$e^{At} = T e^{\Lambda t} T^{-1}$$

$$e^{\Lambda t} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & e^{-t} & 0 \\ 0 & 0 & e^{-2t} \end{bmatrix}$$

$\lambda_1 = 0, \quad \lambda_2 = -1, \quad \lambda_3 = -2$

$\&$ Λ diagonal matrix

$$T^{-1} = \begin{bmatrix} \omega_1^T \\ \omega_2^T \\ \omega_3^T \end{bmatrix} \text{ left eigenvectors}$$

$$\omega_1^T = \begin{bmatrix} 1/4 & 1/2 & 1/4 \end{bmatrix}$$

$$\omega_2^T = \begin{bmatrix} -1/2 & 0 & 1/2 \end{bmatrix}$$

$$\omega_3^T = \begin{bmatrix} 1/4 & -1/2 & -1/4 \end{bmatrix}$$

$$\therefore \dot{x}(t) = T e^{\Lambda t} T^{-1} x(0)$$

$$= \begin{bmatrix} e^{\lambda_1 t} v_1 & e^{\lambda_2 t} v_2 & e^{\lambda_3 t} v_3 \end{bmatrix} \begin{bmatrix} \omega_1^T \\ \omega_2^T \\ \omega_3^T \end{bmatrix} x(0)$$

$$\dot{x}(t) = \left[\sum_{i=1}^{3} e^{\lambda_i t} v_i \left(\omega_i^T x(0) \right) \right]$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \left(\begin{bmatrix} 1/4 & 1/2 & 1/4 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix} \right) + e^{-t} \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \left(\begin{bmatrix} -1/2 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) \right\}$$

$$+ e^{-2t} \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \left(\begin{bmatrix} \frac{1}{4} & -\frac{1}{2} & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix} \right) \right\}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \left[\frac{1}{4} x_1(0) + \frac{1}{2} x_2(0) + \frac{1}{4} x_3(0) \right] + e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \left[-\frac{1}{2} x_1(0) + \frac{1}{2} x_3(0) \right]$$

$$+ e^{-2t} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \left[\frac{1}{4} x_1(0) - \frac{1}{2} x_2(0) + \frac{1}{4} x_3(0) \right]$$

$$x(t) = \left(\frac{1}{4} x_1(0) + \frac{1}{2} x_2(0) + \frac{1}{4} x_3(0) \right) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \left(-\frac{1}{2} x_1(0) + \frac{1}{2} x_3(0) \right) e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$+ \left(\frac{1}{4} x_1(0) - \frac{1}{2} x_2(0) + \frac{1}{4} x_3(0) \right) e^{-2t} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

as $t \to \infty$, $e^{-t} \to 0$ $\&$ $e^{-2t} \to 0$

$\lim_{t \to \infty} e^{-t} = 0$ $\quad \lim_{t \to \infty} e^{-2t} = 0$

$$\therefore \lim_{t \to \infty} x(t) = \left[\omega_1^T x_0(0) \right] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

So since all car positions $x_1(t), x_2(t), x_3(t)$ converges to exact same value when $t \to \infty$

$$\begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix} x(0) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \bar{x} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$ where $x(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{bmatrix}$

meeting at same location is possible. Answer.

(b) They will meet

$$x_1(\infty) = x_2(\infty) = x_3(\infty) = \frac{1}{4} x_1(0) + \frac{1}{2} x_2(0) + \frac{1}{4} x_3(0)$$

$$\therefore x_{meet} = \frac{x_1(0) + 2x_2(0) + x_3(0)}{4}$$, kind weighted average.

Answer

(c) dominant exp rate as $t \to \infty$,

$$x(t) = \left[\omega_1^T T x(0)\right] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \left[\omega_2^T x(0)\right] e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \left[\omega_3^T x(0)\right] e^{-2t} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

as $t \to \infty$ e^{-2t} decays faster than e^{-t}

$$e(t) = x(t) - \bar{x} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \left[\underbrace{\left(\omega_2^T x(0)\right) e^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}}_{\text{dominant, slowest decay } e^{-t} \text{ for } \lambda_2 = -1} + \underbrace{\omega_3^T x(0)\, e^{-2t} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}}_{\text{faster decay } e^{-2t} \text{ for } \lambda_3 = -2} \right]$$

$\therefore$ will converge to zero with dominant decay rate e^{-t} (slowest)

Answer